

Marking Rubric for Applied #1 Assessment (Week 6)

FIT3143 Parallel Computing © Monash University 2026

APPLIED 1 Marking Rubric

		HD	D	C	P	N
Content (40%)						
Task 1 Network topology	Content Correctness, Completeness, and Accuracy (10%)	Able to summarize, compare and contrast on almost all types of network topologies introduced in Week 5. Able to select and explain with strong and flawless reasons on why a certain network topology is chosen for Task 2. The reasoning process must fully consider the realism of simulating the wireless sensor network (WSN) for a set of interconnected EV charging nodes, without missing details.	Able to compare and contrast at least 4 network topologies. Able to select and explain with strong reasons on why a certain network topology is chosen for Task 2. The reasoning process must at least partially consider the realism of simulating the wireless sensor network (WSN) for a set of interconnected EV charging nodes.	Able to compare and contrast at least 3 network topologies. Able to select and explain with acceptable reasons on why a certain network topology is chosen for Task 2, despite some minor errors.	Able to compare and contrast at least 2 network topologies. Able to select and explain why a certain network topology is chosen for Task 2.	Failed to compare and contrast at least 2 network topologies (or) Unable to select and explain why a certain topology is used for the simulator (or) Presenting/reading from copied text (from text book/ web/AI) on different network topologies without any clear comparisons
Task 2 Architecture Design	Content Correctness, Completeness, and Accuracy (15%)	A simulation model architecture is presented with at least 2 clear diagrams. All the diagrams presented are clear and error free. One of the diagrams must show the static structures (e.g., via UML class diagram) and the other one (e.g., via UML communication diagram) must clearly indicate how the objects/components would interact with each other.	A simulation model architecture is presented with at least 1 clear diagram. All the diagrams presented are clear and error free. One of the diagrams must clearly indicate how the objects/components would interact with each other. The message passing protocol matches with the communication requirements in the specification, without any major errors/omissions.	A simulation model architecture is presented with at least 1 clear diagram. The architecture models the described wireless sensor network (WSN) for a set of interconnected EV charging nodes, without major omissions/errors. The architecture has components to model the base station, charging nodes, and charging ports	A simulation model architecture is presented. The architecture models the described wireless sensor network (WSN) for a set of interconnected EV charging nodes, despite major flaws can be found. The architecture has at least 1 component to model the base station and 1 component to model charging node(s)	Failed to present any materials related to the simulation model architecture (or) The presented architecture is clearly not related to the assessment/Task (or) The architecture fails to model the base station or charging nodes

		The message passing protocol matches with the communication requirements in the specification, without any errors/omissions. The architecture models the described wireless sensor network (WSN) for a set of interconnected EV charging nodes, without any component missing. The architecture design considers both shared memory parallelism and message passing parallelism in a cluster environment to improve parallel speed-up, without noticeable flaws.	The architecture models the described wireless sensor network (WSN) for a set of interconnected EV charging nodes, without any component missing.			
Task 3 Analysis	Content Correctness, Completeness, and Accuracy (15%)	Capable to perform communication analysis on the designed architecture for all types of messages, without any errors/omissions. Complexity theory is also being used extensively to draw theoretical conclusions. Capable to perform correct transmission delay computations by using the correct formula/equation The analysis on transmission delay for increasing number of charging nodes and increasing number of base stations is correct. The presentation contains at least 1 graph to show the transmission delay when the number of charging nodes are increasing.	Capable to perform communication analysis on the designed architecture for all types of messages, without any errors/omissions Capable to perform correct transmission delay computations by using the correct formula/equation The analysis on transmission delay for increasing number of charging nodes and increasing number of base stations is correct. The presentation contains at least 1 graph to show the transmission delay when the number of charging nodes are increasing.. The presentation contains at least 1 graph to show the transmission delay when the	Capable to perform communication analysis on the designed architecture for all types of messages, without any major errors/omissions Capable to perform correct transmission delay computations by using the correct formula/equation The analysis on transmission delay for increasing number of charging nodes and increasing number of base stations is largely correct.	Capable to perform basic communication analysis on the designed architecture Capable to perform correct transmission delay computations by using the correct formula/equation	Failed to present any materials related to communication analysis Completely failed to compute transmission delay

		The presentation contains at least 1 graph to show the transmission delay when the number of base stations are increasing. The presentation also studies transmission delay on the other growing factors (e.g., charging ports, growing number of EVs, higher charging frequency, etc). All the graphs presented are convincing and error-free, with abundant data points.	number of base stations are increasing. All the graphs presented are convincing and error-free, with abundant data points.			
Presentation Quality and Q&A (60%)						
Presentation Quality (Clarity and Organisation) (20%)	The presentation is between 6 to 7 minutes. The presentation is <u>exceptionally</u> clear, well-organised, and easy to follow. The conclusion section summarises the presentation well, and discusses major issues, limitations, and future work. The presentation is concise, precise, and accurate (i.e., materials have been digested, analyzed, and chosen carefully without unnecessary information or details). Students are required to extensively use parallel computing terminologies (instead of general	The presentation is between 6 to 7 minutes. The presentation is clear and organised with minor areas for improvement. The conclusion section summarises the presentation, and discusses major issues, limitations, and future work.	The presentation is generally understandable but has major rooms for improvements.	The presentation is somewhat understandable but lacks either organisation or content flow.	The presentation is largely unclear, poorly organised, and hard to follow.	

	computing terms) at this level.				
Q&A (40%)	Capable of understanding and answering all the questions without external tools. All the answers are correct without any flaws. The answers are concise, precise, and accurate (i.e., fully answer the question without unnecessary information or details). Students are required to answer by using parallel computing terminologies (instead of general computing terms) at this level.	Capable of understanding all the questions and answering all the questions without external tools. The answers are all correct without any flaws.	Capable of understanding all the questions and answering all the questions without external tools. The answers are largely correct, despite some minor flaws.	Capable of understanding all the questions and answering all the questions without external tools. The answers are largely correct, despite some major flaws.	Fails to understand the questions (or) Relies on external tools (e.g., AI, software tools, or online tools) to understand the questions or answer the questions. (or) Answers are completely wrong with major flaws/issues.

Special Consideration & Extensions

- From the special consideration procedure, short extensions are not available for group assessments, in-class assessments (including in-class tests/quizzes and presentations), practical assessments (including laboratories), mid-semester tests, or scheduled final assessments. For these assessment types, students must instead apply for special consideration.
- Unless a special consideration extension is approved, the student group/team will generally receive a 0 mark if they fail to attend the in-class session, fail to demonstrate the work, or fail to submit the required files to Moodle.
- For students with special considerations approved by the special consideration team, an alternative form of assessment must be submitted no later than the newly approved deadline. Students are reminded to read and follow the special consideration and alternate assessment sections in the assessment specification.
- If the extended due date falls after Week 14 and you plan to submit on/after Week 14, students are advised to contact the Chief Examiner and/or the Unit Coordinator, as grades may need to be withheld to allow extra time for marking.
- Students who have received special considerations but failed to submit the required files to Moodle by the newly approved deadline for their alternative assessment will be treated as having made no submission (or as having submitted late, if the file eventually arrives after the newly approved due date). For alternative assessments, late penalties will still apply for late submissions.